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COMPUTER SECURITY WK9

1. SCENARIO ANALYSIS:

A cloud-based IT platform requires a strong network security setup to protect sensitive data, applications, and users from cyber threats. The system can use a firewall to monitor and control incoming and outgoing network traffic based on predefined security rules. A Virtual Private Network (VPN) can provide secure remote access for employees by encrypting their connections. Intrusion Detection and Prevention Systems (IDS/IPS) can identify and block suspicious activities such as unauthorized access and malware attacks. Network segmentation can separate databases, application servers, and user networks to limit the spread of attacks. Multi-factor authentication (MFA) should be implemented to provide an additional layer of protection when users log in. Data transmitted across the network should be encrypted using secure protocols such as TLS. Regular security monitoring, software updates, vulnerability scanning, and automated backups should also be performed. This setup helps maintain confidentiality, integrity, and availability while reducing security risks.

2. Concept Research: Next-Generation Firewalls

A Next-Generation Firewall (NGFW) is an advanced network security system that provides more protection than a traditional firewall. It combines stateful traffic inspection with application awareness, intrusion prevention, user identification, threat intelligence, and advanced malware detection. NGFWs can inspect traffic at the application layer, allowing organizations to identify and control specific applications instead of relying only on IP addresses and ports. They can also detect and block threats such as malware, ransomware, unauthorized access, and network attacks. Some NGFWs provide URL filtering, TLS inspection, and network segmentation. They are widely used in businesses and cloud environments to improve visibility, enforce security policies, and protect sensitive systems and data.

3. Tool Practice: Configuring OpenVPN in VirtualBox

I configured OpenVPN in a VirtualBox environment to understand how VPNs provide secure network communication. First, I created a virtual machine and installed a Linux operating system. I then installed OpenVPN and configured the server with the required certificates, keys, IP addresses, and connection settings. A client configuration file was created and used to connect another virtual machine to the VPN server. After establishing the connection, I tested communication between the virtual machines and confirmed that network traffic was securely routed through the VPN. This practical exercise helped me understand VPN configuration, encryption, authentication, and

secure remote access. It also showed me how OpenVPN can protect data from unauthorized interception.

4. Application Practice: Network Security Policy

A network security policy establishes rules for protecting an organization's network, systems, and data from unauthorized access and cyber threats. The policy requires strong passwords, multi-factor authentication, firewalls, antivirus software, and regular security updates. Employees must only access resources necessary for their jobs and should not share passwords or install unauthorized software. Remote users must connect through a secure VPN, while sensitive information must be encrypted during transmission. Network activity should be monitored regularly to detect suspicious behavior. Security incidents must be reported immediately and investigated by authorized IT staff. Regular backups should also be maintained. The purpose of this policy is to reduce security risks, protect confidential information, maintain system availability, and ensure safe and responsible network usage.



NETWORK SECURITY POLICY

PROTECT • PREVENT • ENSURE CONTINUITY



PURPOSE

To protect the organization's network, computers, systems, and data from unauthorized access, malware, data loss, and cyberattacks.

1 ACCESS CONTROL



- Users must have unique usernames and strong passwords.
- Multi-factor authentication (MFA) must be used for important systems.

2 FIREWALL



- Firewalls must monitor and block unauthorized network traffic.

3 NETWORK MONITORING



- Network activity should be monitored regularly to detect suspicious behavior.

4 DATA PROTECTION



- Sensitive information must be encrypted during storage and transmission.

5 SOFTWARE UPDATES



- Operating systems, applications, and security tools must be updated regularly.

6 ANTIVIRUS



- All computers must have updated antivirus and anti-malware software.

7 WI-FI SECURITY



- Wireless networks must use strong encryption and passwords.

8 BACKUPS



- Important data must be backed up regularly and stored securely.

9 INCIDENT RESPONSE



- Security incidents must be reported immediately and investigated.

10 USER TRAINING



- Employees must receive regular cybersecurity awareness training.



THE PURPOSE OF THIS POLICY

The policy reduces security risks, protects confidential information, prevents unauthorized access, and ensures that the organization can continue operating safely during cyber incidents.



KEY BENEFITS

- ✓ Stronger security and risk reduction
- ✓ Protection of sensitive data
- ✓ Prevention of unauthorized access
- ✓ Business continuity and reliability